Supplementary Material
RareAgentBench: A Multi-Pillar, Contamination-Aware Benchmark for Rare-Disease Diagnostic Agents

Anonymous submission

This document is the supplementary material for *RareAgent-Bench*. It is self-contained: each appendix restates the setup it needs, so it can be read without the main paper. It documents (A) the data layers and their construction, (B) the backbone configurations, (C) the agent fairness matrix and adapter patches, (D) per-baseline reproduction against published claims, (E) the temporal-holdout protocol, (F) the pre-registered hypotheses and their family-wise test results, (G) the full cost accounting, (H) a de-leaked MIMIC-IV note probe with a 24-cell agent matrix, and (I) bootstrap and reproducibility notes. Unless stated otherwise, the primary metric is variant-aware Recall@1 (R@1) with an *attempted* denominator: every empty output, parser error, and timeout is counted as a miss, never dropped. Disease predictions are matched by exact identifier, then accepted OMIM–ORPHA mappings, then Orphanet name/synonym match with a RapidFuzz threshold of 90.

A Data Layers and Construction

RareAgentBench draws 84,450 case instances from five sources, organized into four layers that vary evidence structure, clinical realism, scale, and contamination risk. Table 1 summarizes the layers as evaluated in this version.

Table 1: Data layers. "Input" is the evidence an agent receives; "Gold" is the reference label source. Counts are released-pool sizes; per-experiment evaluated N is reported per cell in the main paper.

Layer	Source (count)	Input	Gold
L1	Phenopacket Store (10,051)	curated HPO + variants	OMIM/ORPHA
L1	RareBench-HF (1,122)	sparse HPO list	ORPHA
L2	MIMIC-IV notes, de-leaked (416)	masked discharge summary	ICD→ORPHA (exact)
L3	RareArena RDS (72,661)	free-text vignette	ORPHA
L4	PMC-OA holdout (200)	post-cutoff case text	ORPHA (adjudicated)

L1 Phenotype backbone. Phenopacket Store contributes standardized cases with curated HPO terms and, where available, structured variants; RareBench-HF contributes 1,122 cases for comparability with established rare-disease LLM evaluation. These are the HPO-input layers on which classical/offline baselines (LIRICAL, VC-RDAgent) can run, since those systems require a structured phenotype list.

L2 Real-EHR noise. A de-leaked discharge-note probe over MIMIC-IV v3.1, frozen at 416 cases / 68 diseases (sha256 `67d156aa...`). Candidate admissions are identified through ICD-10–ORPHA mappings, excluding Orphadata categories marked non-rare in Europe. Because MIMIC-IV is credentialed PhysioNet data under a Data Use Agreement, no row-level text or identifiers are redistributed; only aggregate metrics and hashes leave the authorized machine. A de-leaked note-based probe built from this source is documented in Appendix H.

L3 Free-text scale. RareArena contributes 49,760 screening and 22,901 confirmation task instances derived from PubMed Central case reports. We count them as task instances (not unique patients) because the two tasks expose different evidence views.

L4 Temporal holdout. See Appendix E for the full post-cutoff extraction and adjudication protocol.

B Backbone Configurations

Every LLM-based system is evaluated, where technically applicable, with four backbones spanning the observed cost–capability range. Requests are routed through a common OpenAI-compatible gateway to give a single API surface, billing record, and generation log. We use dated model identifiers, not moving "latest" aliases. Table 2 lists the configurations.

Table 2: Backbones evaluated, with pricing and reasoning configuration. Primary comparisons use the lowest reasoning setting each endpoint reliably supports, to reduce backbone-side test-time compute as a confound when measuring scaffold effects and to keep multi-call evaluation tractable.

Role	Model	Price ($/M in / out)	Reasoning
Open-cheap	DeepSeek V4-Flash	0.28 / 0.42	default light
Open-frontier	DeepSeek V4-Pro	0.55 / 2.19	disabled
Mid (primary)	Gemini 3 Flash	0.50 / 3.00	thinking off
Frontier	GPT-5	1.25 / 10.00	`effort=minimal`

Settings held constant across all (agent, backbone) cells: temperature 0.0; seed 42 where the SDK exposes one; per-call timeout 600–1200 s; a retry policy with exponential backoff capped at three attempts; and `max_tokens` at each adapter's published default (2K–6K).

Constrained-reasoning regime. Pilot compatibility tests showed that hidden reasoning could exhaust the output budget of short-turn adapters, producing empty content, parser errors, or timeouts. For the frozen primary runs, GPT-5 uses `reasoning_effort=minimal` and V4-Pro uses reasoning disabled—the only tested configurations that consistently returned final-answer content within adapter limits. All conclusions about GPT-5 and V4-Pro apply specifically to this constrained-reasoning regime; a reasoning-enabled single-LLM control is examined in Appendix F (H6). Failure handling is fixed before aggregation: empty outputs, parser errors, and timeouts count as unsuccessful attempts and remain in the denominator; a cell is marked unavailable only when the adapter–backbone pair cannot execute the required interaction protocol.

C Agent Fairness Matrix and Adapters

The eight diagnostic systems have mutually incompatible dependencies. Each runs in an isolated runtime behind a common adapter interface that receives a canonical case, exposes only the evidence permitted by the current capability and pass, converts it to the system's native input, invokes the isolated process, and returns a prediction log with ranked diagnoses, extracted HPO terms, traces, token usage, cost, latency, and a status code. Adapter modifications are restricted to transport, input projection, output parsing, and instrumentation—never to prompts, tool-selection policies, stopping rules, retrieval corpora, or orchestration logic. Table 3 records each system's native input, adapter strategy, per-case LLM calls, and license.

Audit properties and state isolation. Every run receipt records the canonical case identifier, adapter version, model identifier, configuration hash, subprocess command, random seed, and termination status. A common logging layer aggregates per-call token usage, latency, and billed cost into a case-level receipt; API costs are exact when returned by the gateway and are otherwise estimated from token counts and versioned prices (labeled accordingly, $\leq 5\%$ band). Process isolation contains dependency, parser, retrieval, and state failures within the affected system. Interpreter startup adds roughly 0.5–2 s per case, measured separately from end-to-end latency; system-specific output directories are cleared between cases to prevent cross-case state leakage.

Root-caused failure modes. Two instrumentation issues were found and fixed. (1) MAI-DxO's panel conversation history was not surfaced to the LLM-judge in the first pass, producing parse failures; the corrected pipeline captures the full transcript. (2) MAI-DxO's diagnosis extractor used a greedy regex that could read a discharge-summary vitals line (e.g. "87/35, 97% on RA") as a confidence-0.97 diagnosis and trigger an early-stop short circuit; a vitals filter restores normal multi-round debate. Both fixes and their replay receipts are versioned with the release.

D Per-Baseline Reproduction

Before integrating each system we reproduced its published headline against a small pilot, to confirm the adapter preserves native behavior. Table 4 reports the outcome. Six of nine systems reproduce within ± 5 points; the two that underperform (MAI-DxO, DeepRare-P2) do so for a documented input-distribution reason—both were built for narrative-rich or VCF-bearing inputs and are here given HPO lists or brief vignettes—rather than an implementation error.

The DeepRare P2$\rightarrow$P3 lift ($+20$ points from adding a structured variant block) matches the same lift obtained by a plain single-LLM control given the same variants, which is why we report the genotype-channel benefit as general rather than DeepRare-specific.

E Temporal-Holdout Protocol

The L4 holdout targets contamination by drawing cases published after the backbones' training cutoff. From 2,401 post-cutoff PubMed Central Open Access candidates we extract candidate diagnoses and phenotypes with a fixed pipeline, map them to Orphanet, and manually retain 200 cases satisfying four criteria: a definitive diagnosis; faithful phenotype extraction; verified publication after the prespecified cutoff; and confirmation that the target condition is rare. The holdout is evaluated once, under the frozen agent set, matching rules, and analysis plan; no holdout result is used to revise prompts, adapters, thresholds, or sampling.

Difficulty-matched pre/post comparison. A naive post-cutoff score is confounded by difficulty, so we build a pre-cutoff set with the identical pipeline (same source, query, extraction, mapping, and gold-verification), changing only the publication window. On the clean-gold subset, pooled single-pass R@1 does not drop across the cutoff; if anything it is marginally higher on post-cutoff cases, consistent with newer reports being slightly clearer. We therefore report "no detectable post-cutoff degradation" after difficulty matching, rather than a claim of contamination-freeness: the holdout is not guaranteed disjoint from every development source, so it bounds contamination from a second angle rather than eliminating it.

F Pre-Registered Hypotheses and Family-Wise Tests

Before temporal-holdout unblinding we froze hypotheses H1–H11, the ablation set, the staged sampling budget, cell-level stopping rules, primary metrics, the Holm–Bonferroni correction, and the LLM-judge agreement protocol. The frozen protocol, a timestamped repository commit, and an anonymized read-only registration accompany the release. Table 5 lists the hypotheses; Table 6 reports the family-wise test over the testable subset.

Five of the six testable hypotheses survive the correction. H10 (faithfulness–accuracy decoupling) is reported as exploratory and judge-dependent: on identical repaired traces the faithfulness$\leftrightarrow$ accuracy Spearman correlation is $\rho = 0.457$ under a same-family (Gemini) judge and $\rho = 0.640$ under a cross-family (Claude) judge, straddling

Table 3: Agent fairness matrix. "Calls/case" is the number of backbone invocations per case. Any deviation forced by an unavailable endpoint is declared here; no prompts or orchestration logic were altered.

System	Native input	Adapter strategy	Calls/case	License
MDAgents	MCQA prompt	rank-top-5 reformulation; regex-parse moderator; remove hard-coded model	7–47	None
MedAgents	MCQA prompt	bypass MCQA runner; 3 experts + Chief synthesis	$\sim$10	None
AgentClinic	OSCE dialogue	synthetic OSCE loop; 2nd call for ranks 2–5	$\sim$45	MIT
MAI-DxO	Panel orchestration	router + variant factory; $\geq$2 iterations	10–50	MIT port
DeepRare	Phenopacket + text	local-embedding, no-web environment	20–40	CC BY-NC 4.0
RDMA	EHR free-text	entity extraction; Pillar 1 only	1–3	None
VC-RDAgent	HPO list	offline Stage 1 (IC + Poincaré + freq-LR); Stage 2 opt-in	0 (Stage 1)	None
LIRICAL	GA4GH phenopacket	subprocess `lirical phenopacket`; parse TSV	0	Apache 2.0

Table 4: Per-baseline reproduction on an n=50 pilot. "Ours" is our pilot point estimate; "Paper" is the originally reported figure on that system's own evaluation set. Discrepancies are attributed to input-distribution mismatch, documented per row.

System	Ours (pilot)	Paper	Note
MDAgents	R@1 0.34	0.31–0.39	rank-top-5 reformulation; $\pm$5 pt
MedAgents	R@1 0.36	0.32	3 experts + Chief; $\pm$5 pt
AgentClinic	R@1 0.30	0.28	synthetic OSCE; $\pm$5 pt
MAI-DxO	R@1 0.22	0.45	paper input = NEJM narratives; ours = HPO list
DeepRare (P2)	R@1 0.22	0.71	paper input includes VCF; P2 excludes variants
DeepRare (P3)	R@1 0.42	0.71	structured-variant block, web disabled
RDMA	F1 0.39	0.42	Pillar-1 extraction only
VC-RDAgent	R@1 0.28	0.27	Stage-1 offline, 0 LLM calls
LIRICAL	R@1 0.40	$\sim$0.42	pilot; 0.47 on the full frozen matrix

the pre-registered $\rho < 0.5$ boundary. An earlier $\rho = 0.098$ vs. 0.616 contrast was a trace-capture artifact (the same-family judge had scored truncated traces) and is withdrawn.

G Cost Accounting

The full V1 evaluation comprises 90,046 successfully scored predictions at a total API cost of $270.74 (offline baselines at $0). Table 7 gives per-backbone spend. GPT-5 minimal dominates cost (60% of total) without a consistent accuracy advantage; V4-Flash is more than an order of magnitude cheaper per case.

H De-Leaked MIMIC-IV Note Probe

The MIMIC-IV layer in its original form fed ICD-derived synthetic vignettes in which the ICD long-title effectively named the answer. To remove this leakage we built a note-based probe: the input is the *presentation* portion of a discharge summary (chief complaint, history, exam, labs, imaging), truncated before any diagnosis-revealing section, with the gold disease name masked. The gold label remains code-derived (ICD$\rightarrow$Orphanet exact mapping), which we flag as a code-supervision caveat; this probe is the MIMIC-N column of Table 1 in the main paper.

Subset construction. The deterministic funnel is: 150,033 admissions with any ICD–ORPHA mapping $\rightarrow$ 107,144 exact-mapped only (dropping 37,872 narrower-to-broader and 5,017 broader-to-narrower approximate maps) $\rightarrow$ 18,392 genuinely rare gold $\rightarrow$ 2,325 where the rare disease is the admission's principal diagnosis $\rightarrow$ 1,255 with a usable discharge summary (117 diseases, sha256 `f5d9b437...`) $\rightarrow$ 1,083 after dropping 172 prior-known mentions (112 diseases) $\rightarrow$ 855 in which the gold name never appears verbatim (class A) $\rightarrow$ 687 after the strict Orphanet synonym/abbreviation screen (105 diseases) $\rightarrow$ 416 restricted to diseases carrying Orphanet disease-level HPO annotation (68 diseases, sha256 `67d156aa...`), the frozen scored set. Prior-known detection uses a $\pm$45-character window around each `[MASKED_DIAGNOSIS]` token for history cues ("history of", "hx", "known", "s/p", "diagnosed with"). The synonym screen matters: 19.6% of the nominally clean class-A notes still contained a gold synonym or abbreviation that the crossmap-aware scorer credits as a hit. Truncation alone left the gold name verbatim in only 2,776/78,166 (3.55%) of notes, all subsequently masked; residual gold in `model_input` is 0 by programmatic assertion over the full set. The funnel is itself a finding: only $\sim$1,255 admissions have "a true rare disease as the principal diagnosis with a usable note," quantifying the validity limit of ICD-derived gold. A parallel capped line ($\leq$10 cases per disease, 359 cases / 105 diseases) is used where distribution balance matters more than sample size; 416 and 359 are both subsets of the 687 uncapped strict-A set.

Leakage quantification (before/after). Running an unscaffolded LLM on the same 416 cases twice—once on the full, un-truncated, un-masked note (BEFORE) and once on the de-leaked presentation (AFTER)—isolates the leakage. R@1 drops on all four backbones: V4-Pro $0.500 \rightarrow 0.365$ (-27%), V4-Flash $0.553 \rightarrow 0.353$ (-36%), GPT-5 $0.425 \rightarrow 0.363$ (-15%), Gemini $0.452 \rightarrow 0.373$ (-18%); confidence intervals do not overlap. Weaker models lose more, so leak-

Table 5: Pre-registered hypotheses. H3/H5/H6/H9/H11 are not part of the confirmatory family-wise test reported in Table 6: H3 is a bounded-difference (equivalence-style) check, H5/H9 require data layers deferred in this version (Chinese PUMCH corpus; pedigree-bearing family cases), H6 is a descriptive reasoning-mode ablation, and H11 was retired.

#	Statement	Test
H1	classical $>$ scaffolded LLM on super-rare ($<$1/10^6)	2-prop z
H2	genotype channel (P3) $\geq +10$ pp over P2	paired z
H3	post- vs pre-cutoff R@1 differ by $\leq$5 pp	diff of props
H4	scaffolding helps more on $\geq$4-organ vs $\leq$1-organ cases	diff-in-diff
H5	Chinese (PUMCH) R@1 $\geq$5 pp below English (deferred)	paired z
H6	GPT-5 `medium` better calibrated than `minimal`	paired ECE
H7	cross-agent specialty rank $\rho \geq 0.6$	Spearman ρ
H8	inverted-U: R@1 at 16–30 HPO terms $>$ at $\leq$5	2-prop z
H9	family-aware $\geq +10$ pp on AR/XL cases (deferred)	paired z
H10	faithfulness– and accuracy-rank decouple ($\rho < 0.5$)	Spearman ρ

Table 6: Holm–Bonferroni family-wise correction (α=.05, one-sided in the predicted direction) over the six testable hypotheses. Statistic is a z-value for H1/H8/H2/H4 and a Spearman ρ for H7/H10.

#	Statistic	Holm-adj. p	Survives
H1	$z = 17.5$	2.2e−68	yes
H8	$z = 12.6$	7.8e−36	yes
H2	$z = 6.4$	3.0e−10	yes
H7	$\rho = 0.92$	1.6e−03	yes
H4	$z = 2.6$	9.0e−03	yes
H10	$\rho = 0.35$	3.7e−02	no (exploratory)

Table 7: Cumulative cost by backbone. "†" = cost estimated from token counts ($\leq$5% band). Offline/classical baselines incur no API cost.

Backbone	Cells	Cases (ok)	Total $
GPT-5 minimal	15	20,683	163.50
Gemini Flash	18	23,389	80.48
DS V4-Pro	18	20,666	19.95
DS V4-Flash	16	20,401	6.80
LIRICAL†	2	3,122	0.00
VC-RDAgent†	2	1,785	0.00
Total	71	90,046	270.74

age not only inflates absolute scores but compresses and reorders the model ranking—in the leaked version V4-Flash spuriously "beats" GPT-5.

24-cell agent matrix (de-leaked). On the de-leaked 416-case probe we ran six agent families across four backbones (Table 8, mean micro-R@1 over backbones). The scaffolding picture from the main HPO datasets replicates: the unscaffolded control ties the best light-coordination scaffolds (MedAgents, MDAgents) within overlapping intervals and far exceeds the heavier-orchestration systems, which collapse. This is the second dataset behind the main-paper scaffolding figure. Every one of the 24 cells is complete at 416/416; with the denominator fixed at 416, all failures, timeouts, parser errors and empty-but-successful responses count as misses (1,118 of 9,984 samples, 11.2%). Per-backbone cells are in Table 1 of the main paper.

Table 8: De-leaked MIMIC-IV note probe: mean micro-R@1 over four backbones (denominator 416; all failures counted as misses). The MAI-DxO row uses the regex-fixed rerun; final scoring is unified across all cells.

Agent	Mean R@1
LLM control (no scaffold)	0.364
MDAgents	0.367
MedAgents	0.350
AgentClinic	0.140
DeepRare	0.011
MAI-DxO	0.003

Honest boundaries. Gold is code-derived, not independent clinical adjudication; note coverage is partial, so a selection bias between admissions with and without a discharge summary cannot be excluded; and an organ-system stratification here is disease-level (from the gold disease's Orphanet phenotype list), not case-level, so it aligns structurally but not identically with the other datasets; the 271 cases / 37 diseases dropped at the HPO step are an annotation-coverage artifact of Orphanet `en_product4` rather than a quality filter, but they do narrow disease diversity. Finally, the screen is name-based and case-level: a disease may remain inferable from a characteristic treatment or laboratory pattern. These caveats bound what the MIMIC-N column supports.

I Bootstrap and Reproducibility Notes

All reported point estimates ship with bootstrap 95% confidence intervals (case-level resampling; percentile method). Several small per-agent and per-backbone gaps fall within overlapping intervals and are not treated as definitive rankings. Every confirmatory cell ships a reproducibility receipt containing the run identifier, request identifiers when available, configuration hash, denominator, failure breakdown, and cost. For LLM-judge evaluation of free-text traces, we replay identical frozen traces across judges from different model families, report judge-conditioned results rather than merged scores, and chunk traces longer than 5,000 characters into overlapping 3,000-character windows with 500-character overlap to prevent silent truncation. Deterministic scoring is primary wherever a verifiable target exists; subjective trace scores are reported as exploratory and, for consequential claims, audited against a human-rated subset.